


# UNIVERSITY OF THE ARMED FORCES - ESPE

## Computer Science Department

Object Oriented Programming

36863

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**WorkShop15-Collections**

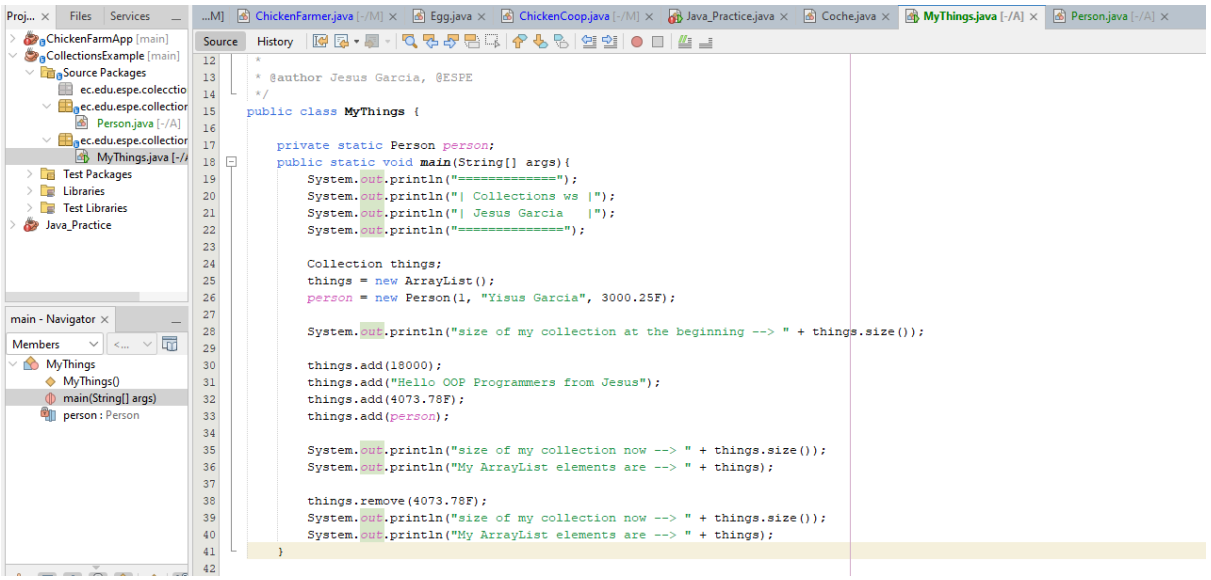
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LASCANO, JORGE EDISON

October 22, 2026

# Evidence

## Java:



```
*
* @author Jesus Garcia, @ESPE
*/
public class MyThings {

    private static Person person;
    public static void main(String[] args){
        System.out.println("=============");
        System.out.println("| Collections ws |");
        System.out.println("| Jesus Garcia   |");
        System.out.println("==============");

        Collection things;
        things = new ArrayList();
        person = new Person(1, "Yisus Garcia", 3000.25F);

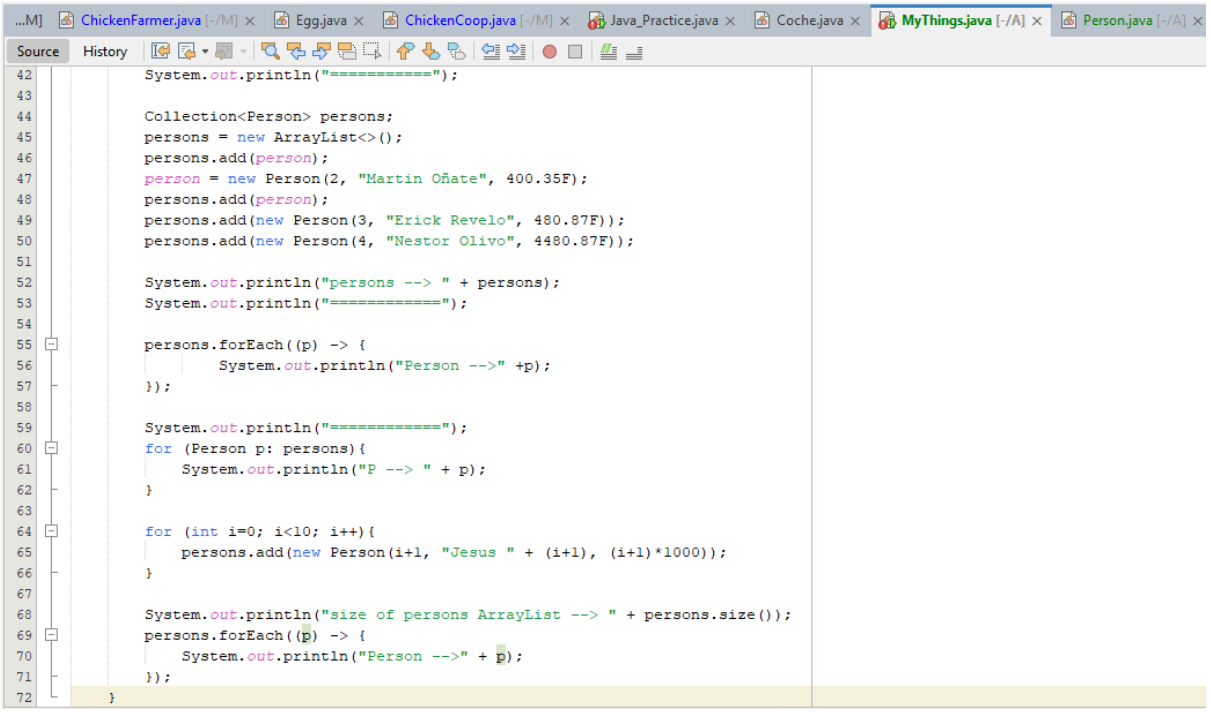
        System.out.println("size of my collection at the beginning --> " + things.size());

        things.add(18000);
        things.add("Hello OOP Programmers from Jesus");
        things.add(4073.78F);
        things.add(person);

        System.out.println("size of my collection now --> " + things.size());
        System.out.println("My ArrayList elements are --> " + things);

        things.remove(4073.78F);
        System.out.println("size of my collection now --> " + things.size());
        System.out.println("My ArrayList elements are --> " + things);
    }
```

```
Output - CollectionsExample (run)
run:
=============
| Collections ws |
| Jesus Garcia   |
==============
size of my collection at the beginning --> 0
size of my collection now --> 4
My ArrayList elements are --> [18000, Hello OOP Programmers from Jesus, 4073.78, Person{id=1, name=Yisus Garcia, salary=3000.25}]
size of my collection now --> 3
My ArrayList elements are --> [18000, Hello OOP Programmers from Jesus, Person{id=1, name=Yisus Garcia, salary=3000.25}]
BUILD SUCCESSFUL (total time: 2 seconds)
```



```
        System.out.println("============");

        Collection<Person> persons;
        persons = new ArrayList<>();
        persons.add(person);
        person = new Person(2, "Martin Oñate", 400.35F);
        persons.add(person);
        persons.add(new Person(3, "Erick Revelo", 480.87F));
        persons.add(new Person(4, "Nestor Olivo", 4480.87F));

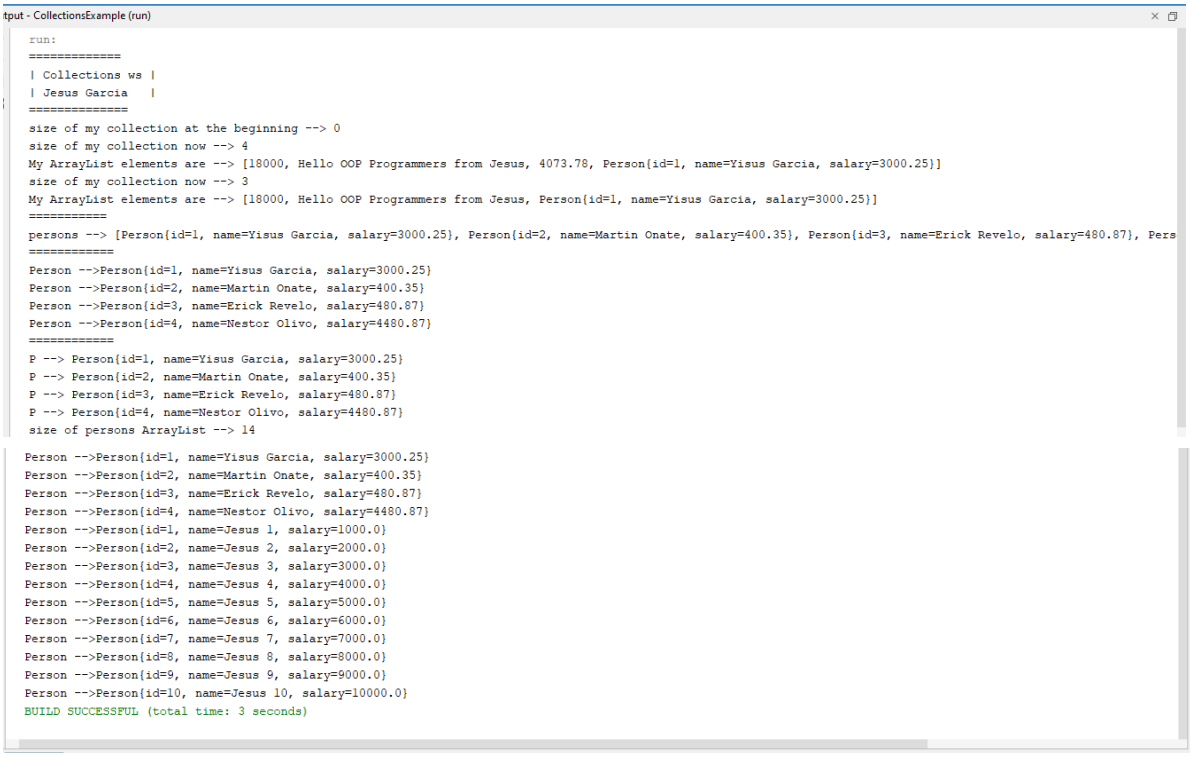
        System.out.println("persons --> " + persons);
        System.out.println("============");

        persons.forEach((p) -> {
                System.out.println("Person -->" +p);
        });

        System.out.println("============");
        for (Person p: persons){
            System.out.println("P --> " + p);
        }

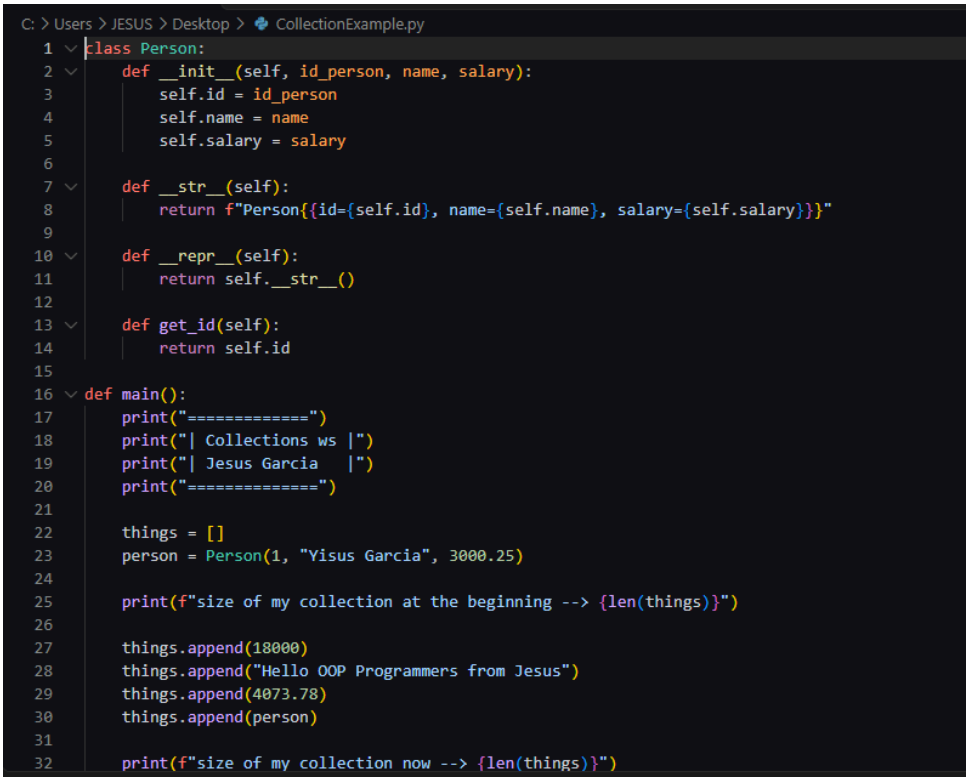
        for (int i=0; i<10; i++){
            persons.add(new Person(i+1, "Jesus " + (i+1), (i+1)*1000));
        }

        System.out.println("size of persons ArrayList --> " + persons.size());
        persons.forEach((p) -> {
            System.out.println("Person -->" + p);
        });
    }
```



```
tput - CollectionsExample (run)
run:
=============
| Collections ws |
| Jesus Garcia   |
==============
size of my collection at the beginning --> 0
size of my collection now --> 4
My ArrayList elements are --> [18000, Hello OOP Programmers from Jesus, 4073.78, Person{id=1, name=Yisus Garcia, salary=3000.25}]
size of my collection now --> 3
My ArrayList elements are --> [18000, Hello OOP Programmers from Jesus, Person{id=1, name=Yisus Garcia, salary=3000.25}]
===========
persons --> [Person{id=1, name=Yisus Garcia, salary=3000.25}, Person{id=2, name=Martin Onate, salary=400.35}, Person{id=3, name=Erick Revelo, salary=480.87}, Pers
============
Person -->Person{id=1, name=Yisus Garcia, salary=3000.25}
Person -->Person{id=2, name=Martin Onate, salary=400.35}
Person -->Person{id=3, name=Erick Revelo, salary=480.87}
Person -->Person{id=4, name=Nestor Olivo, salary=4480.87}
============
P --> Person{id=1, name=Yisus Garcia, salary=3000.25}
P --> Person{id=2, name=Martin Onate, salary=400.35}
P --> Person{id=3, name=Erick Revelo, salary=480.87}
P --> Person{id=4, name=Nestor Olivo, salary=4480.87}
size of persons ArrayList --> 14
Person -->Person{id=1, name=Yisus Garcia, salary=3000.25}
Person -->Person{id=2, name=Martin Onate, salary=400.35}
Person -->Person{id=3, name=Erick Revelo, salary=480.87}
Person -->Person{id=4, name=Nestor Olivo, salary=4480.87}
Person -->Person{id=1, name=Jesus 1, salary=1000.0}
Person -->Person{id=2, name=Jesus 2, salary=2000.0}
Person -->Person{id=3, name=Jesus 3, salary=3000.0}
Person -->Person{id=4, name=Jesus 4, salary=4000.0}
Person -->Person{id=5, name=Jesus 5, salary=5000.0}
Person -->Person{id=6, name=Jesus 6, salary=6000.0}
Person -->Person{id=7, name=Jesus 7, salary=7000.0}
Person -->Person{id=8, name=Jesus 8, salary=8000.0}
Person -->Person{id=9, name=Jesus 9, salary=9000.0}
Person -->Person{id=10, name=Jesus 10, salary=10000.0}
BUILD SUCCESSFUL (total time: 3 seconds)
```

## Python:



```
C: > Users > JESUS > Desktop > CollectionExample.py
class Person:
    def __init__(self, id_person, name, salary):
        self.id = id_person
        self.name = name
        self.salary = salary

    def __str__(self):
        return f"Person{{id={self.id}, name={self.name}, salary={self.salary}}}"

    def __repr__(self):
        return self.__str__()

    def get_id(self):
        return self.id

def main():
    print("=============")
    print("| Collections ws |")
    print("| Jesus Garcia   |")
    print("==============")

    things = []
    person = Person(1, "Yisus Garcia", 3000.25)

    print(f"size of my collection at the beginning --> {len(things)}")

    things.append(18000)
    things.append("Hello OOP Programmers from Jesus")
    things.append(4073.78)
    things.append(person)

    print(f"size of my collection now --> {len(things)}")
```

C: > Users > JESUS > Desktop > CollectionExample.py

```python
def main():
    print(f"My ArrayList elements are --> {things}")

    things.remove(4073.78)
    print(f"size of my collection now --> {len(things)}")
    print(f"My ArrayList elements are --> {things}")

    print("===========")

    persons = []
    persons.append(person)
    person = Person(2, "Martin Onate", 400.35)
    persons.append(person)
    persons.append(Person(3, "Erick Revelo", 480.87))
    persons.append(Person(4, "Nestor Olivo", 4480.87))

    print(f"persons --> {persons}")
    print("============")

    for p in persons:
        print(f"Person -->{p}")

    print("============")

    for p in persons:
        print(f"P --> {p}")

    for i in range(10):
        persons.append(Person(i + 1, f"Jesus {i + 1}", (i + 1) * 1000))

    print(f"size of persons ArrayList --> {len(persons)}")
    for p in persons:
```

## JavaScript:

C: > Users > JESUS > Desktop > CollectionExample.js > Person

```javascript
class Person {
    constructor(id, name, salary) {
        this.id = id;
        this.name = name;
        this.salary = salary;
    }

    toString() {
        return `Person{id=${this.id}, name=${this.name}, salary=${this.salary}}`;
    }

    getId() {
        return this.id;
    }
}

function main() {
    console.log("=============");
    console.log("| Collections ws |");
    console.log("| Jesus Garcia   |");
    console.log("==============");

    let things = [];
    let person = new Person(1, "Yisus Garcia", 3000.25);

    console.log("size of my collection at the beginning --> " + things.length);

    things.push(18000);
    things.push("Hello OOP Programmers from Jesus");
    things.push(4073.78);
    things.push(person);

```

C: > Users > JESUS > Desktop > JS CollectionExample.js > Person

```
17  function main() {
33      console.log("size of my collection now --> " + things.length);
34      console.log("My ArrayList elements are --> [" + things.map(t => t.toString()).join(", ") + "]");
35
36      const indexToRemove = things.indexOf(4073.78);
37      if (indexToRemove > -1) {
38          things.splice(indexToRemove, 1);
39      }
40
41      console.log("size of my collection now --> " + things.length);
42      console.log("My ArrayList elements are --> [" + things.map(t => t.toString()).join(", ") + "]");
43
44      console.log("==========");
45
46      let persons = [];
47      persons.push(person);
48      person = new Person(2, "Martin Onate", 400.35);
49      persons.push(person);
50      persons.push(new Person(3, "Erick Revelo", 480.87));
51      persons.push(new Person(4, "Nestor Olivo", 4480.87));
52
53      console.log("persons --> [" + persons.map(p => p.toString()).join(", ") + "]");
54      console.log("==========");
55
56      persons.forEach((p) => {
57          console.log("Person -->" + p.toString());
58      });
59
60      console.log("==========");
61
62      for (let p of persons) {
63          console.log("P --> " + p.toString());
```